Autoshaping models summary

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Model breakdown

To assess how drug dose, sex, and response types varied over the course of an autoshaping procedure, we constructed a Bayesian *hurdle negative binomial* model. This model captures both the likelihood of making a particular response on a given trial (*P(Response)*, like a logistic model) and the likelihood of making x number of responses on that trial, given a response was made (*P(Number of responses)*, like a Poisson model).

Details and specifications

Outcome

Left and right lever/head magazine entries were aggregated within trials to create total sums of lever presses and head entries during the trial period. The outcome is therefore the **integer count of responses** during the trial.

Predictors

Models were fit as full factorial models with the following predictor variables:

- **Response type** (categorical): whether response being counted is head entry or lever press
- **Drug** (continuous): dose in mg/L of methamphetamine
- **Trial** (continuous): trial of autoshaping procedure across all sessions/days
- **Sex** (categorical): whether individual was male or female

Drug was standardized on the grand mean, and Trial was standardized on individual-level means. Response type and Sex were effect coded.

Random effects

To account for individuals and potentially cohorts/litters selectively performing particular strategies (e.g., sign-tracking and goal-tracking), we allowed **individual and cohort level responses to vary** across response type and trial (along with their interaction). This is the maximal random effects structure for this model.

Priors

All fixed effect weights (for both the hurdle and count portions of the model) were estimated with weakly informed priors (*Normal*(0, 3)) and random effect variances were estimated with weakly regularizing priors (*Exponential*(1)).

Sampling

Models were fit with the `brms` package using the `cmdstanr` HMC engine. Parameter posteriors were estimated with 12,000 samples (4,000 burn-in) across 4 chains. Chains demonstrated **acceptable convergence**, verified with a visual analysis and supported by all parameters having Rhat values below 1.01.

Results